

Generation Interconnection System Impact Study Report

For Queue Project Q20-1632

Stent 138 kV

May 04, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1632. The Interconnection Customer proposes a 54.6 MW battery energy storage facility interconnecting at the Stent 138 kV substation in the Pennsylvania area. The request was received on December 25, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 518 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,094,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Pico to Stent 138.0 kV: 117.8% loading (224 MVA) following the outage of the Big Swan to Owatonna 345.0 kV; DFAX = 50.1%.

Barbadoes to Stent 138.0 kV: 109.4% loading (212 MVA) following the outage of the Big Swan to Owatonna 345.0 kV; DFAX = 49.9%.

The Fruitland to St. Jons 138.0 kV is overloaded to 103.2% of its 233 MVA emergency rating for the loss of the Tombstone to Marcus Hook Energy Lp 500.0 kV L. The project contributes 10.5% DFAX (5.7 MW) to this overload.

St. Johns to Hathaway St 138.0 kV: 101.5% loading (273 MVA) following the outage of the Marcus Hook Energy Lp to Tombstone 500.0 kV Line; DFAX = 10.5%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pico to Stent 138.0 kV	Rebuild line with higher-capacity conductor	\$820,000	22% (\$182,000)
Stent to Barbadoes 138.0 kV	Reconductor line	\$1,138,000	22% (\$252,000)
Fruitland to St. Johns 138.0 kV	Reconductor line	\$889,000	38% (\$336,000)
Hathaway St to St. Johns 138.0 kV	Reconductor line	\$857,000	38% (\$324,000)
	Total allocated to project		\$1,094,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.